THE SCHEME THEORETIC IMAGE OF A PROPER MORPHISM INTO A SEPARATED FINITE TYPE SPACE IS PROPER

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-morphisms.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The statement includes the displayed triangle and its two numbered hypotheses; the proof uses density and closedness of the image to get surjectivity onto Z and then properness of Z over B.

Editorial note. The five references of the counted spans are recorded as non-load-bearing navigation pointers; the argument that consumes them stays inside the counted proof span.

Editorial note. Preamble: the three native excerpt spans of sources/stacks-spaces-morphisms/preamble.tex (lines 8-13, 14-18 and 154-217). The chapter file itself carries no preamble text.

Allowed standard prerequisites (editorial).

- The construction of the scheme theoretic image. Section section-scheme-theoretic-image, used at proof lines 8250-8252. Fixes what Z is.
- Quasi-compactness of the morphism h and density of the image in Z. Lemma lemma-quasi-compact-quasi-separated-permanence and Lemma lemma-quasi-compact-scheme-theoretic-image, used at proof lines 8253-8256. Gives quasi-compactness of h and density of |h|(|X|) in |Z|.
- The image of X in Y is closed. Lemma lemma-universally-closed-permanence, used at proof lines 8257-8259. Makes |h|(|X|) closed, hence equal to |Z| up to density, so X -> Z is surjective.
- The image of a proper morphism is proper. Lemma lemma-image-proper-is-proper, used at proof lines 8259-8260. Gives properness of Z -> B.

Source reference. Original lemma 08AJ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 10534–10534. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 073B: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9055–9055. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 082Z: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 10269–10269. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 04NX: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6095–6095. Navigation pointer into the Stacks

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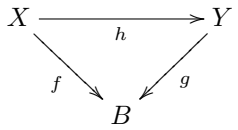
Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original section 082W: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 10266–10266. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Lemma 0.2. *Let S be a scheme. Let*



be a commutative diagram of morphism of algebraic spaces over S. Assume

(1) *$X \to B$ is a proper morphism,*
(2) *$Y \to B$ is separated and locally of finite type,*

Then the scheme theoretic image $Z \subset Y$ of h is proper over B and $X \to Z$ is surjective.

Proof. The scheme theoretic image of h is constructed in Section 082W. Observe that h is quasi-compact (Lemma 073B) hence $|h|(|X|) \subset |Z|$ is dense (Lemma 082Z). On the other hand $|h|(|X|)$ is closed in $|Y|$ (Lemma 04NX) hence $X \to Z$ is surjective. Thus $Z \to B$ is a proper (Lemma 08AJ). □